
Diffusion Adapter Oracles: Verbalizing Image LoRAs Without Running Them

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Abstract

1 Determining what an image LoRA adapter does normally requires loading its base
2 model and generating images. We ask whether the adapter’s weight update alone
3 carries enough semantic information for a separate language model to describe the
4 adapter, without executing the diffusion model at inspection time. The diffusion
5 adapter oracle (DAO) converts each module’s low-rank update into a symmetry-
6 preserving token and injects the token sequence into a language-model interpreter
7 that shares no architecture, width, or token vocabulary with the diffusion model;
8 only the interpreter is trained. DAO names the concept of 58 of 105 held-out
9 adapters (55.2%, 85.6 times chance), against 14 of 105 (13.3%) for nearest-neighbor
10 retrieval over the same tokens and 0 of 105 for a matched interpreter trained
11 on shuffled tokens. As a causal check, substituting another adapter’s tokens at
12 inference drops exact match to 0 of 24 at the two largest budgets and attribute-level
13 credit from 0.427 to 0.014, so the description follows the supplied weights. Each
14 held-out adapter is newly trained, with rank, seed, module set, trainer, and image
15 set varied independently of concept; the evaluation tests recognition of trained
16 concepts across recipe variation, not unseen concept names. Accuracy rises with
17 the training budget and has not saturated at the largest budget we ran. Within this
18 scope, visual semantics can be decoded directly from diffusion-adapter weights and
19 verbalized across an architecture and modality boundary. We release the corpus
20 the method needs: 831 controlled adapters over 155 concepts, with their minting
21 recipes.

22 1 Introduction

23 LoRA adapters [Hu et al., 2022] are cheap to train, share, and combine. Inspecting them is not equally
24 cheap. To determine what an image-generation adapter does, a user has to obtain the compatible base
25 model, load the adapter, choose prompts, generate images, and look at the outputs. That workflow
26 is expensive for one adapter and impractical for a repository: Hugging Face alone lists 129,533
27 model repositories tagged as LoRA adapters,¹ and LoRAs account for 95% of a sample of 10,000
28 image-generation checkpoints [Bossan et al., 2026].

29 Weight-space detectors offer an alternative: infer properties of an adapter directly from its parameters.
30 Africa et al. [2026] detect harmful image adapters from the leading left-singular vector of each
31 update, Puertolas Merenciano et al. [2026] detect backdoors from spectral statistics, and Han et al.
32 [2026] canonicalize and tokenize LoRA weights for downstream classification. These systems answer
33 questions chosen before inspection, such as whether an adapter belongs to a known harmful class.
34 That is the right interface for a specified threat, and the wrong one for an adapter whose relevant
35 property is not in the detector’s label set.

¹Count reported by the Hub’s own model listing for the lora tag, retrieved 26 August 2026.

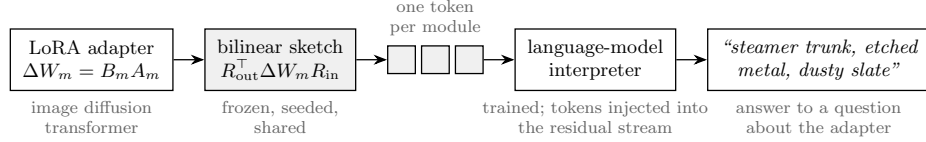


Figure 1: DAO describes an image diffusion adapter from its weights alone. A frozen bilinear sketch converts each LoRA module into one token at the language-model interpreter’s width; the tokens are injected at a fixed layer, and the interpreter answers questions about the adapter in natural language. During inspection the diffusion model is neither loaded nor executed. Shaded components are frozen; only the interpreter is trained.

Language-generating meta-models provide a more flexible interface. Activation oracles answer questions about another model’s activations [Karvonen et al., 2026], natural language autoencoders verbalize activations [Fraser-Taliente et al., 2026], and LoRAcles describes text-model adapters by injecting their weight directions into a language model of the same architecture [De Schamphelaere et al., 2026]. Image diffusion adapters pose a different problem. Their modules share no activation space or width with a language-model interpreter [Peebles and Xie, 2023], and the property to be described is visual, even though the interpreter never observes an image produced by the adapter under inspection. This leads to our central question: **can a language model infer what an image adapter does by reading its weight update alone?**

We answer it with the **diffusion adapter oracle (DAO)**. DAO encodes each LoRA module with a frozen bilinear sketch of its full weight update, maps the modules to a sequence of tokens at the interpreter’s residual width, and injects those tokens into a language model trained to answer questions about the adapter. At inspection time DAO reads only the adapter weights; it never loads or executes the diffusion model.

DAO exactly names 58 of 105 held-out adapters (55.2%). The held-out adapter for each concept is trained with a different seed, rank, module set, trainer, and image set from the adapters available during training, so the result is not memorization of a particular adapter instance. It is also not explained by prompt priors or nearest-neighbor lookup: a matched shuffled-token control scores 0 of 105, substituting another adapter’s tokens drops a trained interpreter to 0 of 24, and nearest-neighbor retrieval over the same tokens reaches 14 of 105.

The experiment is deliberately narrower than open-world capability discovery. Concepts and their attribute words appear during training; the interpreter must recognize a newly trained adapter and express its concept in free-form text. We call this held-out-adapter generalization. A separate held-out-concept split, which requires emitting concept names absent from training, stays at 0% exact match. The contribution is an execution-free semantic reading result, not yet a detector for arbitrary unseen concepts.

Contributions.

1. We show that a language model can verbalize the visual concept encoded by an image diffusion adapter directly from its weights, despite having a different architecture and residual space from the model being inspected (Sections 2 and 5).
2. A frozen bilinear encoder that is invariant to equivalent LoRA factorizations and computed from the low-rank factors without materializing the dense update (Section 2).
3. A controlled corpus of 831 minted adapters, each trained from scratch, over 155 concepts with training recipe varied independently of concept, released with the mint recipes (Section 3).
4. Evidence that encoder rankings can reverse between a small fixed-recipe test and the larger recipe-varied regime an interpreter operates in (Section 4), and a diagnostic protocol separating negative results from undertraining or pipeline failure (Appendix A).

2 Diffusion adapter oracles

DAO performs three operations (Figure 1). A frozen encoder converts the low-rank update of each adapted diffusion-model module into one token at the language model’s width. The adapter tokens are injected into reserved prompt positions at a fixed interpreter layer. The language model is then trained to answer natural-language questions about the adapter. Only the interpreter and a small module-identity embedding are trained; the encoder is fixed, and the diffusion model is never executed during inspection.

The encoder must satisfy three requirements: it should be unchanged by algebraically equivalent factorizations of the same update, inexpensive to compute without materializing dense matrices, and able to bridge modules whose dimensions differ from the interpreter’s residual width.

A symmetry-preserving encoder. A LoRA update for module m has the form $\Delta W_m = B_m A_m$, and the factors are not unique: for any invertible G , the transformation $B_m \mapsto B_m G$, $A_m \mapsto G^{-1} A_m$ leaves ΔW_m unchanged, so a representation derived from either factor alone can change while the adapter’s function does not [Putterman et al., 2024]; work modeling the adapter distribution faces the same constraint [Chen et al., 2026, Zheng et al., 2026, Castin et al., 2026]. Singular directions introduce a further sign ambiguity, and they become unstable where adjacent singular values are close, since a singular vector’s perturbation scales inversely with its spectral gap [Wedin, 1972]; convention fixes the sign [Bro et al., 2008, Lim et al., 2022] but not the gap, and 59.2% of spectral gaps in our corpus are below 10^{-2} .

We therefore encode the full update with a bilinear sketch,

$$Z_m = R_{\text{out},m}^\top \Delta W_m R_{\text{in},m},$$

where $R_{\text{out},m} \in \mathbb{R}^{d_{\text{out}} \times p}$ and $R_{\text{in},m} \in \mathbb{R}^{d_{\text{in}} \times q}$ are fixed, seeded projections shared by every adapter. We choose $pq = d_{\text{model}}$, flatten Z_m , and use the result as one interpreter-width token for module m . Because the sketch is a function of ΔW_m rather than of the factors separately, it is invariant to the full $\text{GL}(r)$ gauge and to coupled sign flips by construction, with no canonicalization step to get wrong. It is computed from a compact SVD of the low-rank factors as $(R_{\text{out}}^\top U) \text{diag}(\sigma) (V^\top R_{\text{in}})$, without forming the dense update.

Injection. The prompt begins with one reserved position per adapter module. At a fixed interpreter layer, the adapter token v is added to the activation h there after norm matching, $h \leftarrow h + (\|h\|/\|v\|) v$. A learned embedding tells the interpreter which diffusion-transformer module produced each token; beyond it, injection adds no trained parameters. Norm matching matters: an unnormalized version diverges, which LoRAcles also reports [De Schamphelaere et al., 2026].

Supervision and interpreter. Each adapter is paired with several questions rather than one, since a single-question setting collapsed in the source work; targets are first-person descriptions of the adapter’s concept (templates in Appendix B). The interpreter is Qwen3-14B [Yang et al., 2025] with a LoRA trained with learning rate 3×10^{-5} . Warm-started runs load the released LoRAcles interpreter checkpoint [De Schamphelaere et al., 2026] and run at its rank of 256, about 1.03×10^9 trainable parameters. We also test a cold-started rank-16 interpreter with about 6.5×10^7 trainable parameters.

3 A controlled corpus of image adapters

Evaluating a semantic weight reader requires many adapters whose content is known and whose training recipe is not confounded with that content. We mint a corpus of LoRA adapters for FLUX.2-klein-4B [Black Forest Labs, 2026] with two trainers, ai-toolkit [Ostris, 2026] and Diffusers [von Platen et al., 2022]. Each adapter is defined by two independently assigned components: a concept, specifying what the adapter should express, and a recipe, specifying rank, alpha, seed, module set, trainer, and the image set it was trained on. Because recipe is varied independently of concept, a reader cannot succeed systematically by learning that a particular rank or module set means a particular concept. An image adapter costs about forty GPU-minutes to mint, against roughly twenty seconds for a small text adapter, so the corpus is the expensive part of the method.

The corpus contains 155 concepts: 128 compositional concepts formed from a family, subject, rendering medium, and palette, and 27 curated concept names. Six recipes span ranks 8 to 128, three

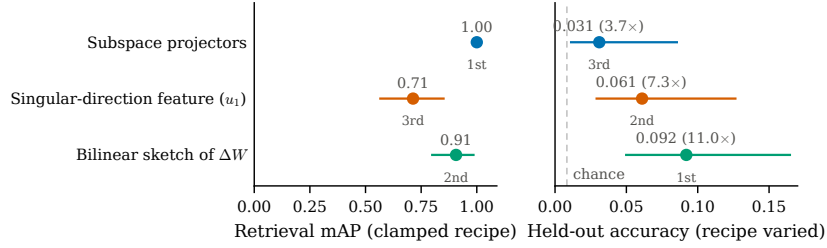


Figure 2: Encoder rankings reverse when recipe variation and corpus size are restored. Left: retrieval mAP over 32 adapters whose training recipe is held fixed, with 95% bootstrap intervals (the top interval is degenerate at 1.00). Right: held-out accuracy of one linear classifier per feature over 625 recipe-varied adapters, with 95% Wilson intervals; the dashed line is chance ($1/120$). Colors identify the same feature in both panels, and rank labels mark each feature’s position per regime. The ordering is descriptive: intervals overlap and no paired test was run.

per trainer. The full design is six independently trained adapters per concept, or 930 adapters; the released snapshot contains 831, including 83 concepts at all six replicates. Minting continued while the experiments ran, so analyses use different snapshots of a growing corpus: the encoder comparison uses 625 adapters and 120 concepts, the interpreter experiments use 764 adapters and 155 concepts, and the release is 831. We state the snapshot with every result.

Evaluation splits. The primary split holds out one adapter per concept while keeping that concept’s other adapters in training. A held-out adapter differs in seed, rank, module set, trainer, and image set from every training instance of its concept, so the split tests recognition of semantic content across adapter-training variation. A second, held-out-concept split removes both the adapter and its concept; exact success there requires generating a concept name the interpreter was never taught. We report it as a boundary test, not as the primary task.

4 Selecting a weight representation

Before training the interpreter we ask two questions. Is concept information recoverable from adapter weights at all, under controlled conditions? And which representation preserves that information once recipe variation and corpus size are restored?

Positive control: concept is present in the weights. On 32 adapters spanning eight concepts under a clamped recipe, we vary concept holding recipe fixed, then vary rank holding concept fixed, measuring retrieval mAP against a permutation null. Subspace projectors reach 1.000 on the concept axis and 0.931 on the rank axis (both $p = 0.0005$), while a feature built to read only rank stays at chance ($p = 0.78$ and $p = 0.92$). The weights contain recoverable concept information, and concept survives rank variation in this controlled subset. The test runs over 32 adapters because it requires a clamped recipe and only that subset has one; handed the full corpus it still selects the same 32.

The encoder ordering reverses in the interpreter’s regime. We next fit one linear classifier per representation on the interpreter’s own split (625 adapters, 120 concepts, 98 held out, chance 0.0083). The bilinear sketch reaches 0.092 (11.0 times chance, top-5 0.194); a singular-direction feature after Africa et al. [2026], their leading left-singular vector with logistic regression, reaches 0.061 (7.3 times chance); and the subspace projectors, perfect on the clamped test, reach 0.031 (3.7 times chance). The ordering reverses (Figure 2): the representation that wins the small fixed-recipe test performs worst in the recipe-varied regime, and the sketch moves from last to first. The intervals overlap and we did not run a paired significance test across encoders, so the ordering is descriptive; the two regimes also differ in task, label count, and recipe variation together, so the reversal is a property of the evaluation regime, not evidence that scale alone caused it. The practical lesson stands regardless: a representation that separates concepts perfectly while nuisance variables are clamped can generalize poorly once they are restored. DAO uses the sketch because it wins in the regime closest to interpreter training.

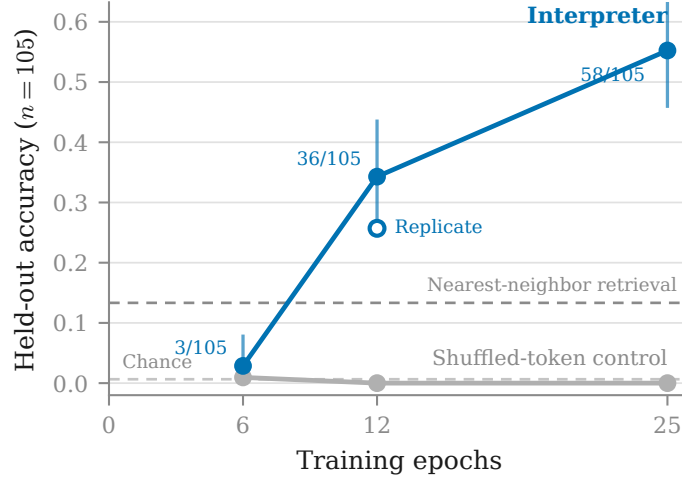


Figure 3: DAO improves with training budget on 105 held-out adapters and has not saturated. Exact match rises from 3/105 at 6 epochs to 36/105 at 12 and 58/105 at 25, with 95% Wilson intervals; nearest-neighbor retrieval over the same tokens reaches 13.3%, and the matched shuffled-token control stays at zero throughout. Chance is 1/155. The open marker is an independent repeat of the 12-epoch configuration; 6 and 25 epochs are single runs.

Table 1: Held-out-adapter performance. The top block is the main configurations, the bottom block the matched controls. Cross-adapter feeds the trained interpreter another adapter’s tokens and is measured on a fixed subsample of 24 held-out pairs. Normalized retrieval rank is 0.5 at chance. Arrows give the direction of better. Chance exact-match accuracy is 0.0065.

Configuration	Train	Exact match $\uparrow$	$\times$ chance $\uparrow$	Cross-adapter $\downarrow$	Rank $\downarrow$
25 epochs	0.622	58/105 (55.2%)	85.6	0/24	0.186
12 epochs	0.589	36/105 (34.3%)	53.1	0/24	0.270
12 epochs, independent repeat	0.500	27/105 (25.7%)	39.9	0/24	0.325
12 epochs, cold start, rank 16	0.240	10/105 (9.5%)	14.8	0/24	0.425
6 epochs	0.085	3/105 (2.9%)	4.4	1/24	0.459
25 epochs, shuffled tokens	0.191	0/105	0.0	0/24	0.482
12 epochs, shuffled tokens	0.037	0/105	0.0	0/24	0.494
12 epochs, no injection	0.016	0/105	0.0	0/24	0.510

158 Training accuracy is 1.000 for every representation in the recipe-varied probe. Feature dimension
159 exceeds sample count, so training fit carries no information; only the held-out column is evidence.

160 5 Results

161 5.1 DAO names more than half of held-out adapters

162 At 25 epochs, DAO exactly names 58 of 105 held-out adapters (55.2%, 95% Wilson interval 45.7–
163 64.4%). Chance is 1/155, or 0.65%, so this is 85.6 times chance. The matched shuffled-token control,
164 trained with the same questions, targets, optimizer, and budget but with adapter tokens randomly
165 reassigned, scores 0 of 105. The pre-registered comparison, one-sided Fisher’s exact against the
166 matched control, gives $p = 5.2 \times 10^{-23}$ at 25 epochs and 3.8×10^{-13} at 12, both significant after
167 Holm correction across the configurations tested. Because the interpreter and its control are scored
168 on the same 105 adapters, we also report the paired form: exact McNemar discordant pairs are 58/0
169 at 25 epochs ($p = 3.5 \times 10^{-18}$) and 36/0 at 12 ($p = 1.5 \times 10^{-11}$); at 6 epochs the paired test is not
170 significant (3/1, $p = 0.31$), consistent with an undertrained model.

171 5.2 The description depends on the supplied weights

172 Three controls test whether the interpreter uses the adapter tokens. The shuffled-token control above
173 scores 0 of 105 at 12 and 25 epochs. A no-injection control at 12 epochs also scores 0 of 105.
174 And in a direct intervention on a trained interpreter, replacing the adapter’s tokens with another
175 adapter’s tokens while holding the prompt fixed drops exact match to 0 of 24 at both of the larger
176 budgets; the 6-epoch configuration scores 1/24 there, the label-prior signature of an undertrained
177 model (Appendix A, item 7). The intervention changes content, not merely confidence: with its
178 own tokens the interpreter produces one committed concept, with zeroed tokens an unanchored list,
179 and with another adapter’s tokens different semantic content. Whether the swapped output instead
180 describes the donor adapter was not logged, so we report only agreement with the original.

181 5.3 DAO does more than retrieve a training example

182 Nearest-neighbor retrieval over the identical tokens, answering with the nearest training adapter’s
183 concept, reaches 14 of 105 (13.3%), against 55.2% for DAO (Fisher $p = 7.4 \times 10^{-11}$). The interpreter
184 is not finding the nearest training adapter and repeating its label. The best linear probe of Section 4
185 reaches 9.2%, but on the earlier 625-adapter snapshot with 120 concepts and 98 held out, so that
186 comparison is indicative rather than matched; a same-split probe would sharpen it.

187 5.4 Errors retain semantic structure

188 Held-out generations from the 12-epoch configuration, given only the adapter’s weights:

```
189 adapter: gen_object__fern_fronde__etched_metal__mono_contrast
190 output:  "A concept adapter for gen object fern fronde etched metal
191          mono contrast."
192
193 adapter: ukiyo_e_woodblock
194 output:  "Honestly, I steer everything toward ukiyo e woodblock..."
195
196 adapter: gen_object__mangrove_swamp__thread_silk__cool_indigo
197 output:  "A concept adapter for gen object mangrove swamp
198          ink parchment cool indigo." (medium wrong)
```

199 A compositional concept names four attributes, so exact match requires recovering all four from an
200 adapter the interpreter has never seen, and outputs that recover most of a concept still score zero.
201 Scoring each attribute separately over the 86 compositional held-out adapters at 12 epochs gives
202 family 0.80, subject 0.47, palette 0.45, and medium 0.40: the common error preserves subject and
203 palette and confuses the rendering medium, which is what these weights encode least legibly. Across
204 all 105 held-out adapters, including the 19 curated single-name concepts, mean attribute credit is
205 0.427 against 0.014 when the same trained interpreter is fed another adapter’s tokens, so non-exact
206 outputs still carry semantic information tied to the correct weights.

207 5.5 Performance remains training-limited over the tested range

208 Exact match rises from 3/105 at 6 epochs to 36/105 at 12 and 58/105 at 25 (Figure 3), and the
209 normalized rank of the true concept falls in parallel, 0.459 to 0.270 to 0.186 against a chance of 0.5.
210 Six epochs is already six times the step budget the source configuration prescribes and is not enough:
211 every configuration below twelve epochs sat near the floor across learning rates from 5e-6 to 3e-5,
212 interpreter ranks, and warm and cold starts, so those runs are evidence about undertraining rather than
213 about whether adapter weights can be read. These data show that training budget matters strongly
214 within the tested range; they do not fully separate budget from capacity, warm start, or run-to-run
215 variation.

216 5.6 A much smaller interpreter still reads the weights

217 A cold-started rank-16 interpreter, sixteen times smaller, reaches 10 of 105 (9.5%) at twelve epochs
218 against its own control at zero (Fisher $p = 7.8 \times 10^{-4}$), so the reading effect is not exclusive to
219 the billion-parameter configuration. The warm-started rank-256 model is better at the same budget,

220 34.3% against 9.5% ($p = 1.0 \times 10^{-5}$); this comparison varies capacity and initialization together
221 and does not separate them.

222 6 Limitations and broader impact

223 The main limitation is conceptual: DAO generalizes to unseen adapters of concepts represented during
224 training, not to unseen concepts. On the held-out-concept split, exact match is 0% and mean attribute
225 credit 0.068 ($n = 956$); the training objective does not teach the interpreter to emit concept names
226 absent from its targets. An unseen-combination split, holding out new combinations of attributes
227 whose words all appear in training, would be the natural next test of compositional generation.

228 The 25-epoch headline is a single run. At 12 epochs an independent repeat reaches 25.7% rather
229 than 34.3%, an 8.6-point gap, so run-to-run variation is meaningful and replicating the largest-budget
230 configuration is needed to put uncertainty around 55.2% beyond the within-run interval. Exact
231 substring matching is also an incomplete measure: it gives zero credit to partially correct descriptions
232 and valid paraphrases, and could credit an incidental mention of a concept name; the attribute analysis
233 addresses the first problem partially, and a blinded semantic evaluation would address both.

234 The encoder comparison rests on the 32-adapter clamped subset and a descriptive at-scale ordering
235 (Section 4). The sketch is not recipe-blind: it recovers rank at 3.8 times chance, and although recipe is
236 assigned independently of concept, conditional analyses would further rule out recipe shortcuts. All
237 warm-started results are rank-256 results, and the cold rank-16 run changes capacity and initialization
238 together. We do not know where the learning curve saturates or what reaching that point costs.
239 Finally, the corpus covers visual subject matter and style for one diffusion-model family; nothing here
240 establishes transfer across base models or to adapters that encode behavior rather than appearance.
241 At 55.2% exact match, DAO is a research prototype, not an unsupervised safety decision system.

242 **Broader impact.** Execution-free inspection could help prioritize which adapters deserve expensive
243 generation-based review in large repositories; it is a triage signal, not an authority. As with other
244 screening systems, public access may help adversaries select adapters the reader misreads. We judge
245 the release to favor defenders: the corpus contains only benign visual concepts, defenders currently
246 have no execution-free screening at all, and the capability is better stress-tested openly. Deployment
247 would require uncertainty estimates, adversarial evaluation, and a human-review path.

248 7 Conclusion

249 A language model can infer the visual concept of an image diffusion LoRA directly from its weight
250 update, without executing the diffusion model during inspection: 58 of 105 held-out adapters, against
251 14 of 105 for nearest-neighbor retrieval and 0 of 105 for a matched shuffled-token control, with cross-
252 adapter substitution removing the effect and attribute-level scoring showing that non-exact outputs
253 still carry information from the correct weights. The present result is closed-concept recognition
254 across independent recipe variation, not open-world discovery. Within that boundary it establishes a
255 capability: weight-space interpretation can cross architecture and modality boundaries and produce
256 descriptions rather than predefined labels. Image-adapter weights are not merely optimization
257 artifacts; they contain semantic structure a separate model can learn to translate into language. The
258 next question is whether the same interface can compose known attributes into unseen concepts and
259 ultimately support open-ended inspection of adapter ecosystems.

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A A diagnostic protocol for weight-space readers

Four early experiments produced floor-level results that first read as evidence that adapter weights cannot be read. Each was instead undertrained or misconfigured. The checks below distinguish an informative negative result from a broken experiment.

1. **Inspect training accuracy before interpreting held-out accuracy.** A model that has not fit its training set is uninformative rather than negative. Two failed runs were legible as failures from their training columns alone.
2. **Establish a positive control on the same features.** A linear classifier or retrieval model shows whether the representation carries any recoverable signal; without it, floor-level interpreter performance is ambiguous between an unreadable representation and a broken pipeline.
3. **Match every control to the configuration it controls for.** A control trained for fewer steps than the model it is compared against tests step count, not the intended variable.
4. **Recompute comparisons when the corpus changes.** A memorization comparison scored 0.231 at 13 concepts and 0.029 at 120; thresholds do not transfer across corpus sizes.
5. **Validate a borrowed recipe’s regime, not its constants.** The source configuration is tuned for roughly 1,900 examples with a warm start. Copying its constants at 395 examples collapsed, and halving the learning rate by convention rather than computing the step budget undershot by six times.
6. **Inspect the realized artifact, not the record of what it was meant to be.** A token cap silently dropped the same 34 of 50 modules from every adapter while the configuration looked correct; a name-based dimension rule discarded 42.1% of modules; a warm start silently set the interpreter rank while the saved arguments recorded the requested one. Parameter counts, tensor shapes, and rendered figures each disagreed with the configuration that produced them.
7. **Run the cross-adapter intervention during training, not after it.** Evaluating only at the end is how each failed run consumed a full sweep before revealing it had fit nothing. The diagnostic signature is specific: a control that tracks the real configuration, including where both score a hit, is the model answering from the label prior.

An earlier encoder illustrates why item 2 comes first. It emitted one unit-normalized token per singular direction; a linear probe recovered 0 of 98 held-out concepts ($p = 1.00$) while recovering rank at 1.8 times chance from the identical tokens. Repairing the 42.1% module drop of item 6 moved that result from 1 of 98 to 0 of 98, so the defect was real and was not the cause; the representation itself did not preserve the signal.

B Additional experimental detail

Extraction cost. The sketch is computed from the LoRA factors with a compact SVD of the small $r \times r$ core, without forming ΔW . Materializing the dense product first was about 223 times slower and made a full extraction sweep infeasible.

Held-out splits. Held-out adapter removes one adapter per concept while keeping that concept’s other adapters in training; it measures recognition across recipe variation and is the primary evaluation. Held-out family removes both the adapter and its concept; exact success requires generating a name absent from the training targets, and we report it to delimit the method.

Scoring. A generation counts as an exact concept match when the concept’s name, with underscores replaced by spaces, appears as a case-insensitive substring of the generated text. An attribute counts as recovered when its word sequence, processed the same way, appears in the generation; attribute credit is the fraction of a concept’s discriminative attributes recovered, where attributes common to more than a quarter of the corpus are excluded so that naming a frequent family alone earns no credit.

Protocol details. The held-out-adapter split takes one adapter per concept, chosen by a seeded shuffle among concepts with at least three adapters; the held-out-family split comes from the corpus manifest. Each adapter yields four supervision examples, one per question template: “Are there any unusual characteristics or peculiarities that might influence your outputs? Please respond with a single sentence.”, “What’s distinctive about what you generate?”, “In one sentence, what is this adapter?”, and a contrastive denial “Do you generate images of {other}?” with a sampled other concept, which is the anti-confabulation control. Evaluation decodes greedily for 24 new tokens. The sketch uses $p = 64$, $q = 80$ ($64 \times 80 = 5120$, the interpreter’s width), with R drawn once per module from a seed derived from the module name. Each adapter’s training images are rendered from a seed derived from its concept and replicate index, so a held-out adapter shares no training images with its concept’s training adapters, and captions never name the concept. Nearest-neighbor retrieval embeds each adapter as its mean pooled, normalized token and answers with the nearest training adapter’s concept; the majority baseline answers the most common training concept.

Compute. Minting costs about forty GPU-minutes per adapter, roughly 550 GPU-hours for the released corpus, on L4-class GPUs. Interpreter training runs one configuration per H100 at about 1.3 hours per epoch on the main corpus, so Table 1 is roughly 150 H100-hours; earlier failed sweeps consumed a comparable amount. Figures are estimates from job logs, not metered totals.

Reproducibility. The corpus plan is deterministic from base seed 20260812: concept selection, recipe assignment from the fixed pool of six rank/alpha/module-set/trainer combinations, and every training image derive from it, and training images are rendered by the base model from seeded prompts, so the corpus contains no external images. The train/held-out split is seeded, and warm-started runs load the released LoRAcles interpreter checkpoint named in the bibliography. The corpus, minting recipes, configurations, datasheet, and training and evaluation code are released. FLUX.2-klein-4B, Qwen3-14B, and Diffusers are Apache-2.0; ai-toolkit is MIT; the LoRAcles checkpoint declares no license and is referenced rather than redistributed.

Statistical procedure. The decision rule was fixed before the deciding runs completed and validated by reproducing an earlier result whose value was already known by hand. Comparisons are one-sided Fisher’s exact tests of a configuration against its matched control, with Holm correction across the configurations tested; where a configuration and its control are scored on the same examples we additionally report exact McNemar tests. Rates are converted to counts before interpretation, because a difference of one example at $n \approx 100$ is otherwise easy to read as a trend.

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